FRONT VIEW — TARGET FORM



~145×72×14 mm · ~220 g · NTS

mesh radio marks shown in blue

WHAT CHANGED VS MONOLITH

— Monolith = flagship research device (titanium spine, optical docks).

— Wrong phone for a rancher in a -35 °C truck cab or a fly-in community.

— ArcticPhone is rural-first: mesh-native, satellite SOS-ready, offline-first, repairable, community-owned.

— Every ArcticPhone IS a node — calls, texts, data ride the community mesh.

ARCHITECTURE OPTIONS (DECISION REQUIRED)

OPT	APPROACH	PROS	CONS
A	Rugged Android + ArcticChain mesh HAT	fast to market, app ecosystem	Android bloat, cert effort
B	Custom Linux + mesh SoC (integrated SX1262)	clean, sovereign, repairable	OS work, smaller ecosystem
C	Hybrid: Linux core + Android layer	best of both	longest dev time

CAPABILITY SET

CAPABILITY	IMPLEMENTATION
Mesh node	SX1262 LoRa 915 + signed envelopes — every phone extends the network
Satellite SOS	L-band messenger module now; own constellation later
Offline apps	maps, manuals, forms, messaging EN/FR — useful with zero bars
Cellular	LTE Cat-1/4 when towers exist — proven path from L1
Power	replaceable battery + USB-C PD + solar trickle option
Rugged	IP68 · -30..+55 °C · glove UI · 1.5 m drop target

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UNITS **mm**

TARGET SPECIFICATION

PARAMETER	TARGET	PARAMETER	TARGET
Envelope	~145×72×14 mm	Mass	~220 g
Display	5–6in transflective	SoC	quad-core, LTE, GNSS
Mesh radio	SX1262 915 + MCU	Sat SOS	L-band module (rev-A)
Battery	replaceable ~4000 mAh	Charging	USB-C PD + solar
OS	Android Go OR Linux	UI	glove-friendly, EN/FR
Rating	IP68 · -30..+55 °C	Repair	standard screws, open back

BOM CATEGORIES (NOT FIXED — ARCHITECTURE PENDING)

CATEGORY	OPTIONS / NOTES
Compute+display	rugged Android module (A) OR custom board (B)
Mesh module	SX1262 + nRF52/ESP32 — same envelope as fleet
Sat messenger	inReach-class module, Iridium Certus OEM, or future own-sat
Cellular	LTE Cat-4 modem + SIM — proven path from L1 HAT
Power	4000 mAh LiFePO4 pouch, USB-C PD, solar pad option
Enclosure	PC+TPU bumpers, gasketed ports, captive screws

ROADMAP TO FIRST UNIT

— Phase 0 (now): architecture decision + industrial design — this sheet.
— Phase 1: prototype on dev-kit + mesh HAT — prove mesh+SOS on existing hardware.
— Phase 2: custom enclosure + integration — print → machine → mold decision.
— Phase 3: certification (ISED/RF) + pilot batch — 50–100 units for founding communities.
— Phase 4: volume — repairable design pays off at scale (modules, not glue).

UNIQUE VS EVERY OTHER PHONE

— The only phone that is a FIRST-CLASS MESH CITIZEN — your phone extends the network.
— Satellite SOS as a core feature, not an accessory — for places towers aren't.
— Offline-first: apps, maps, forms, messaging all useful with zero bars.
— Community-owned: the network your phone joins belongs to your community.
— Repairable by design — a local tech fixes it with hand tools.

RELATIONSHIP TO OTHER NODES

— Phone = personal node; L1 = field/shared; Totem = public; Mite = environmental; Anchor = infrastructure.
— All speak the same signed mesh — a call routes phone→phone→brick→satellite.
— Phone is the highest-volume node — its radio is the network's user terminal.

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THE ONE-LINE PITCH

“A phone that joins YOUR network — not just the carrier's.”

Every ArcticPhone is a mesh node, a repeater, a satellite SOS beacon, and an offline-first computer — in one rugged, repairable, community-owned device.

SIX THINGS NO OTHER PHONE DOES

#	FEATURE	WHY IT MATTERS
1	TRI-RADIO	LoRa mesh + LTE + satellite — three independent paths; at least one always works
2	EVERY PHONE A REPEATER	each unit extends coverage for the next — density IS the network
3	SATELLITE SOS STANDARD	one hardware key press reaches help anywhere on Earth — not an accessory, a core feature
4	OFFLINE DIGNITY SUITE	maps, manuals, forms, messaging, notices — fully useful with zero bars
5	COMMUNITY SYNC	Totem public boards push to your phone; your posts can reach the Totem
6	REPAIRABLE BY DESIGN	screws not glue, standard connectors, a local tech fixes it with hand tools

VS THE ALTERNATIVES

NEED	SMARTPHONE	RUGGED PHONE	SAT MESSENGER	MESH RADIO	ARCTICPHONE
Cellular calls	✓	✓	—	—	✓
Works with no towers	—	—	✓	✓	✓
Satellite SOS	app only	—	✓	—	✓ built-in
Extends a network	—	—	—	✓	✓
Offline apps	partial	partial	—	—	✓ full
Repairable	glue	screws	sealed	—	screws
Community-owned	carrier	carrier	corporate	DIY	community

THE MOAT — WHY IT COMPOUNDS

— Network effect: every phone sold adds a repeater — the 100th user has better coverage than the 1st. Competitors can't copy deployed density.

— Community ownership: the mesh belongs to the town, not a carrier — that's a procurement argument AND a loyalty argument.

— Data stays home: messages route community mesh first — sovereignty is a feature regulators and communities both want.

— Module ecosystem: open bay spec → third parties build sat, thermal, battery modules → platform, not product.

— The phone is the user terminal for the whole ArcticChain stack — mesh, swarm, Totem, satellites. One device, the entire ecosystem.

WHAT IT IS NOT

— NOT a smartphone killer — it's for places smartphones fail.

— NOT a Starlink terminal — L-band messenger class for SOS/short messages.

— NOT a toy mesh radio — it's certified cellular + satellite + mesh, carrier-grade.

— NOT locked to one community — joins any ArcticChain mesh, carries identity.

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UNITS	**mm**

RADIO BLOCKS

MESH — SX1262
LoRa 915 MHz
0.3–50 kbps
5–15 km LOS
~50 mW

CELLULAR — LTE Cat-4
B1/2/5/13/66
50–150 Mbps
tower range
~1–2 W

SATELLITE — L-band
Iridium Certus class
~22 kbps
global
~3–5 W

All three share the SoC via SDIO/UART — coexistence filtering required: cellular fundamental can desense LoRa.

COEXISTENCE & FAILOVER

SCENARIO	RADIO USED	BEHAVIOR
In community mesh range	LoRa primary	calls/texts route peer-to-peer signed
No mesh, cell exists	LTE	normal cellular + mesh backhaul opportunistic
No cell, no mesh	L-band sat	SOS + short messages only (22 kbps)
All three present	mesh→LTE→sat priority	mesh for local, LTE for internet, sat reserved SOS
Simultaneous TX	TDM scheduler	cellular TX pauses LoRa RX windows

LINK BUDGET (LORA, TYPICAL)

PATH	RANGE	DATA	USE
Phone ↔ phone	1–3 km	text/voice-meta	direct mesh call
Phone ↔ Brick/Totem	3–10 km	notices/data	community services
Phone ↔ Anchor	5–15 km	backhaul relay	reach the gateway
Phone ↔ satellite	global	22 kbps	SOS + telemetry only

ANTENNA LAYOUT (TARGET)



GPS/BT: ceramic patch rear

WHY TRI-RADIO WINS

— Single-radio phones fail exactly where they're needed — no towers, no mesh, no service. Three radios = three independent lifelines.

— LoRa mesh is FREE — unlicensed 915 MHz, zero carrier cost, community-owned.

— LTE is familiar — works today, every Canadian knows it.

— L-band sat is the insurance policy — works at the cabin, the fire line, the North. SOS doesn't care about coverage maps.

— Failover is automatic and signed — the network decides, user just talks.

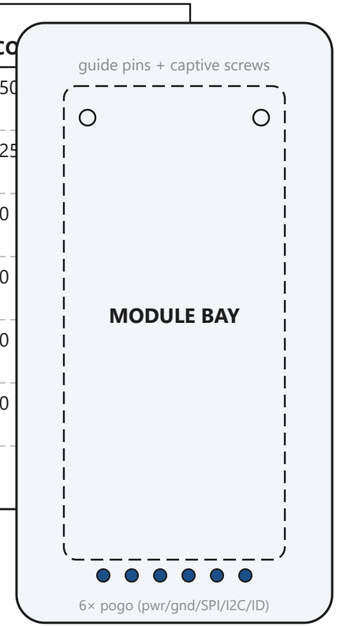
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MODULE BAY SPEC (TARGET)

MODULE	FUNCTION	INTERFACE	POWER	EST. CO
SAT-TERM	full Iridium/Globalstar terminal	USB3 + pogo	5 V 2 A	$80–150
THERMAL	FLIR Lepton-class camera	MIPI + pogo	5 V 1 A	$120–25
EXT-BATT	second 4000 mAh cell	pogo charge	pass-through	$25–40
LORA-BOOST	external antenna + PA	U.FL + pogo	5 V 0.5 A	$15–30
E-INK DOCK	public display accessory	SPI + pogo	5 V 1 A	$40–80
STORAGE	1 TB encrypted card	microSD slot	—	$50–90
CUSTOM	third-party open spec	pogo + USB-C	5 V ≤2 A	open

BAY MECHANICS (REAR VIEW)



hot-swap? NO — power down to change
signed module auth on insert

MODULE AUTH & SAFETY

— Each module carries signed ID — phone verifies before enabling power.
— Power budget enforced: module draws only within its rated envelope.
— Thermal/current foldback — a bad module can't cook the phone.
— Open spec published — third parties can build modules (revenue share).
— Modules are field-swappable with one screwdriver — 60-second change.

ECOSYSTEM PLAY

— Phase 1: first-party modules (sat, thermal, battery).
— Phase 2: certified partners — ag sensors, medical, survey gear.
— Phase 3: open marketplace — community-built modules.
— Every module is a node accessory — extends the mesh's reach.
— This is how a phone becomes a PLATFORM — the moat compounds.

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SOS CHAIN — ONE KEY, EVERY PATH

— Physical SOS key (dedicated, glove-proof, hardware interrupt — not software).
— Press → phone broadcasts signed SOS on mesh → nearest node relays.
— If no mesh ack in 30 s → escalate to LTE data.
— If no LTE → satellite messenger sends GPS + vitals packet.
— Responder sees: identity, GPS, battery, mesh path taken — full provenance.
— Works upside-down, wet, -30 °C, screen cracked — hardware path only.

MESH DASHBOARD (LIVE)

METRIC	WHAT USER SEES
Coverage bubble	your mesh range on a local map — live, offline
Node count	how many phones/bridges repeat your signal right now
Message latency	mesh vs LTE vs sat — which path carried your last msg
Mesh score	your contribution — messages relayed, hours online
Community notices	Totem board content synced to phone

OFFLINE DIGNITY SUITE

APP	OFFLINE CAPABILITY
Maps	full regional maps pre-loaded — routing without bars
Manuals	equipment/service docs — searchable, no data
Forms	government/clinic/ag forms — fill, sign, queue to send
Messaging	mesh SMS between phones — store-and-forward
Notices	community bulletin — Totem content cached
Camera	photos tagged GPS — queue upload when backhaul returns

GLOVE-FIRST HARDWARE

ELEMENT	DESIGN CHOICE
SOS key	dedicated hardware button, 3 mm travel, textured
Display	transflective — readable in sun, low power
UI	high contrast, large targets, haptic confirm
Buttons	3 physical (vol/PAN) + SOS — no touchscreen-only
Audio	loud speaker + mic — works with mask/glove
Charging	USB-C + pogo — one-hand dock charge

COMMUNITY INTEGRATION

— Phone joins community mesh automatically on first boot — no carrier account needed for mesh service.
— Community admin sees node health (opt-in) — coverage gaps visible.
— Totem public boards sync notices to phones within range.
— Your posts can go to the Totem — community bulletin from your pocket.
— Data sovereignty: mesh traffic stays on community infrastructure unless explicitly routed out.
— EN/FR UI throughout — language is a first-class setting.

BATTERY & POWER UX

— Replaceable battery — swap in field with one tool, no adhesive.
— USB-C PD fast charge — 0→80% in ~45 min.
— Solar trickle mode — leaves on dash, gains ~10%/day.
— Ultra-low-power standby — 30 days standby on mesh listen.
— Battery health shown — honest capacity, not fake percentages.

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P · ARCTICPHONE

FINISH & MATERIALS — FINAL LOOK

AF-P-107

REV A · 2026-11-12

APPEARANCE REV A



P ARCTICPHONE — FRONT

~145 × 72 × 14 mm · ~220 g · IP68



REAR — MODULE BAY

bay covers 40% of rear — hot-swap modules

COLOR / FINISH OPTIONS

GRAPHITE
default — professional

FOREST
field/rural identity

HI-VIS
SAR / search roles

MATERIALS SPEC

COMPONENT	MATERIAL	FINISH	NOTE
Chassis	polycarbonate + TPU bumpers	textured matte	IP68 sealed
Display	transflective LCD 5–6in	anti-glare	sunlight + gloves
Buttons	silicone overlay + metal	tactile 3mm	glove-proof
Antenna	LDS laser-direct structuring	—	no external whips needed
Bay cover	PC + TPU gasket	thumb-screw	tool-light swap

DESIGN NOTES

— Rugged but NOT ugly — matte graphite reads 'serious tool', not 'military'.

— Bumper corners absorb 1.5 m drops onto gravel — replaceable.

— Module bay cover is the visual signature — slightly raised, thumb screw.

— Transflective display: readable in sun WITHOUT backlight — days of battery.

— No glass back — polycarbonate shrugs off drops and cold.

— Color options are filament/finish only — same internal BOM.

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